MIN PAING HEIN

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Professional Summary

Innovative Computer Science Major (Year 3) with a specialization in building high-fidelity, “glitch-aesthetic” web platforms that solve complex engineering problems. Proven track record of architecting full-stack systems from AI-powered async processing to static analysis engines and MLOps pipelines. Combining deep algorithmic knowledge (Blind 75) with advanced UI/UX engineering (Framer Motion) to deliver the software that is both performant and immersive.

Education

Expected Graduation: Jul 2027

King Mongkut's University of Technology Thonburi (KMUTT) — *Bangkok, Thailand*

Bachelor of Science in Computer Science

- Relevant Coursework: Data Structures & Algorithms, Web Development, Machine Learning, Object-Oriented Programming (Java).

Technical Skills

Languages: TypeScript, JavaScript (ES6+), Python, SQL, Java, HTML/CSS

Frontend: Next.js (App Router), React, Framer Motion, Tailwind CSS, Recharts, Vite

Backend: Node.js (Express), FastAPI, Python (Pandas/NumPy), RESTful APIs, Puppeteer

Data & AI: Scikit-Learn, Google GenAI (Gemini), AST Parsing, Code Generation, RAG Architectures

Infrastructure: TiDB (Serverless, MySQL), Cloudflare R2, Docker, Linux, Git

Projects

Prima | No-Code ETL & ML Pipeline Generator

Full Stack Engineer (React, FastAPI, Pandas, Scikit-Learn)

- **Engineered a "Glass-Box" Data Engineering platform** that allows users to visually construct complex Scikit-Learn preprocessing pipelines (Imputation, Encoding, Scaling) and export them as production-ready Python code.
- **Developed a Code Transpiler Engine** in Python that converts JSON-based UI state ("Recipes") into syntactically correct, dependency-resolved Python scripts, effectively automating the data cleaning process.
- **Built a high-performance EDA API** using **FastAPI** and **Pandas** to calculate statistical distributions and outliers for large CSVs in milliseconds, employing stateless architecture to minimize server costs.
- **Implemented a "Live Preview" Sandbox** that safely replays transformation steps on sample data, handling edge cases (Infinity/NaN serialization) to provide real-time feedback to the user.
- **Designed a "Dark Glass" aesthetic** with Tailwind CSS and Framer Motion, utilizing Virtualization (IntersectionObserver) to render 50+ interactive histograms without UI lag.

Syzygy | Automated API Drift & Governance Platform

Full Stack Engineer (React, Node.js, GitHub API, Static Analysis)

- **Developed a developer-tooling platform** that detects "API Drift" by analyzing discrepancies between Backend Route Definitions and Frontend API consumption in real-time.
- **Engineered a custom Static Analysis Engine** using **Regex-based** parsing to extract API signatures from multi-language codebases (Python/Flask, Node/Express, Next.js) without runtime execution.
- **Implemented a fuzzy-logic matching algorithm (Levenshtein Distance)** to identify and categorize route mismatches, typos, and orphaned endpoints with high accuracy.
- **Built a secure, in-memory processing pipeline that streams GitHub** repository archives (Zip) for analysis, utilizing AES-256 encryption to secure user credentials at rest.
- **Designed a "Command Center" Dashboard** featuring a cybernetic UI with custom SVG data visualization and an Auto-Remediation Engine that generates copy-pasteable code patches.

Eidolon | AI-Powered Academic Intelligence Suite

Full Stack Engineer (Next.js, TypeScript, Gemini 2.5, Cloudflare R2, TiDB)

- **Architected a long-running async processing pipeline** to convert 3-hour lecture transcripts into structured study notes, utilizing SQL polling to handle Gemini’s 65k+ token generation windows.
- **Engineered a custom PDF Generation Microservice** using Puppeteer on the server-side, capable of rendering syntax-highlighted code blocks, automatic Table of Contents, and academic-grade CSS.
- **Implemented robust Data Integrity patterns using SQL Transactions (ACID compliance)** and optimistic locking to prevent race conditions during concurrent file uploads.
- **Optimized Cloud Infrastructure** by integrating **TiDB Serverless** and **Cloudflare R2 (S3-compatible)**, reducing potential cloud storage costs by ~40%.